

Time Series Analysis and Stock Forecasting

Internship Project Report

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Offered By

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Introduction

Stock market data changes every day, and understanding these changes is important for investors and analysts. Looking at historical stock prices helps us identify trends and make better predictions about future price movements. However, analyzing large amounts of stock data manually is difficult and takes a lot of time.

In this project, I developed an interactive dashboard using Streamlit that allows users to upload historical stock data in CSV format. The dashboard provides different visualizations to explore market trends and uses the Facebook Prophet model to predict future stock prices. This makes it easier to understand stock performance and estimate future trends.

Objective

The main objective of this project is to build an interactive dashboard that can analyze historical stock market data and predict future stock prices. The dashboard is designed to make stock analysis simple, interactive, and easy to understand.

Problem Statement

Stock market datasets are usually large and contain information collected over long periods of time. Analyzing this data manually is not only time consuming but also makes it difficult to identify important trends and patterns. Investors need tools that can visualize historical data and provide future predictions in a simple way. This project solves that problem by combining data visualization and forecasting in one application.

Tools Used

The following tools and technologies were used in this project:

Python

Streamlit

Pandas

NumPy

Plotly

Matplotlib

Facebook Prophet

Scikit Learn

Project Workflow

Data Collection

The user uploads a CSV file containing historical stock market data.

Data Preprocessing

The application checks whether the required columns are available, converts the date column into the correct format, and arranges the data in chronological order.

Exploratory Data Analysis

The dashboard displays basic information about the dataset and shows historical stock price trends to help users understand the data.

Data Visualization

Several interactive charts, including candlestick charts and trend graphs, are used to present stock price movements clearly. These visualizations help users identify patterns and market behavior.

Forecasting

The Facebook Prophet model is trained using historical closing prices. After training, the model predicts future stock prices for the selected forecast period.

Model Evaluation

The forecasting results are evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These metrics help measure the accuracy of the prediction model.

Results

The dashboard successfully analyzes historical stock prices and generates future forecasts. Users can explore price trends, moving averages, and prediction results through interactive charts. The application also provides evaluation metrics to help understand the performance of the forecasting model.

Conclusion

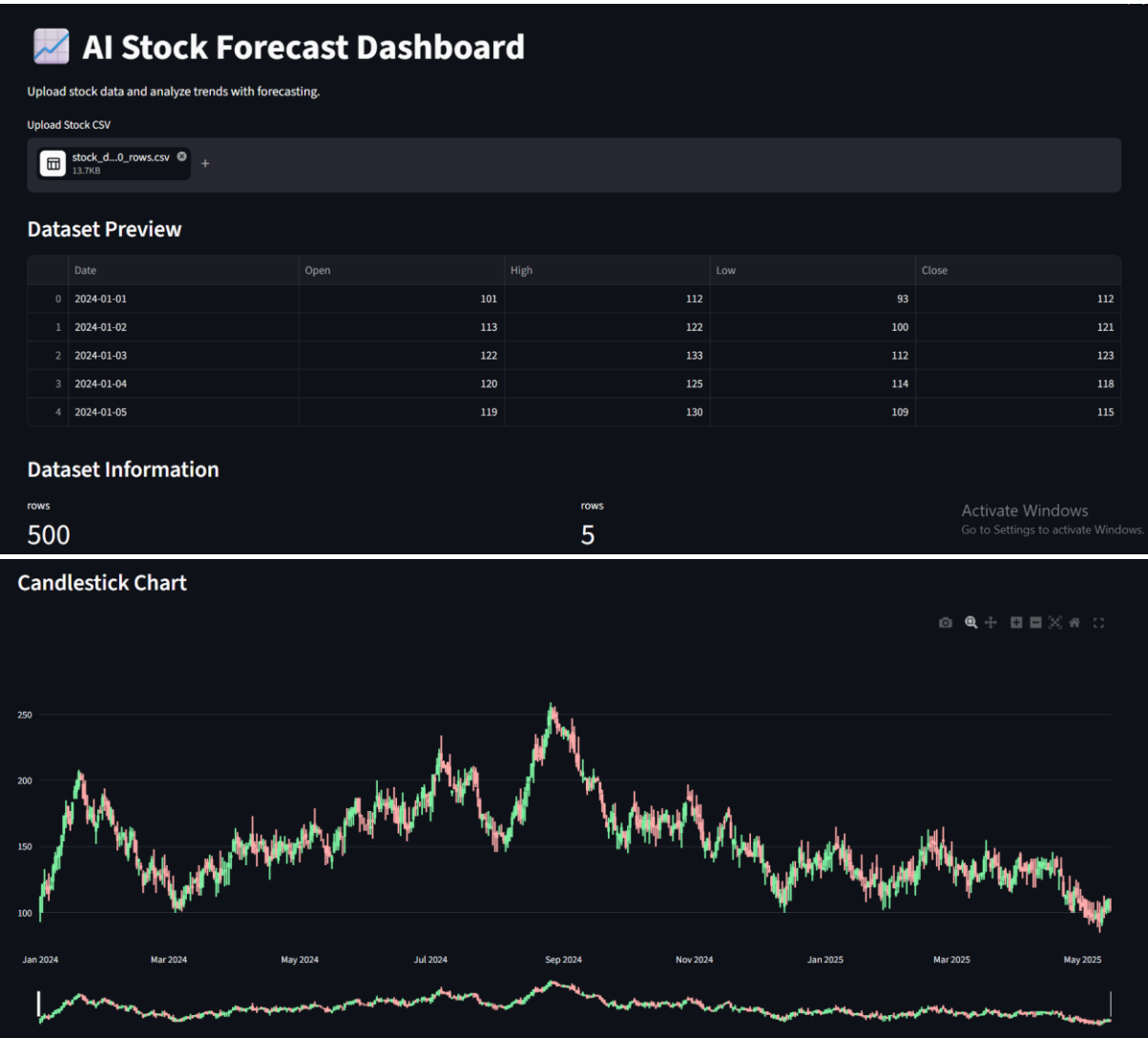
This project helped me understand how time series analysis and machine learning can be applied to real world financial data. It also improved my practical knowledge of data preprocessing, forecasting techniques, visualization, and Streamlit application development. Overall, the project provided valuable hands on experience in building an interactive stock analysis system.

Future Scope

This project can be improved in several ways. Future versions may include LSTM based forecasting models, live stock market data through APIs, additional technical indicators, improved prediction accuracy, and online deployment so users can access the dashboard from anywhere.

Learning Outcomes

- During this project, I learned:
- Time Series Analysis
 - Data preprocessing
 - Financial data visualization
 - Forecasting using Facebook Prophet
 - Moving average analysis
 - Interactive dashboard development with Streamlit
 - Model evaluation using MAE and RMSE



Closing Price Trend



Moving Average Analysis

Select Moving Average Window



Model Accuracy

MAE

13.43

RMSE

17.22

Forecast Data

	ds	yhat	yhat_lower	yhat_upper
500	2025-05-15 00:00:00	114.7993	93.4843	136.3151
501	2025-05-16 00:00:00	114.7537	93.4158	136.1067
502	2025-05-17 00:00:00	114.4264	92.3494	135.669
503	2025-05-18 00:00:00	114.436	92.5583	137.0153
504	2025-05-19 00:00:00	113.1886	90.7726	134.4626
505	2025-05-20 00:00:00	113.4634	90.6099	134.2807
506	2025-05-21 00:00:00	113.004	91.5251	135.2775
507	2025-05-22 00:00:00	113.5134	91.3947	135.3922
508	2025-05-23 00:00:00	113.4678	93.3819	137.1884
509	2025-05-24 00:00:00	113.1405	92.2352	135.8563

